V-21—Air elimination requirements for bulk meters

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1.0 Purpose

The purpose of this Bulletin is to define the policy relating to the installation of a bulk meter without an air eliminator.

2.0 Scope

This bulletin applies to all bulk meters installed at loading racks for the delivery of refined petroleum products.

3.0 References

Section 245 of the *Weights and Measures Regulations* requires an automatic air-vapour eliminator to be installed with all meters except slow-flow meters and slow-flow dispensers.

4.0 Background

In the case of bulk meters installed at loading racks for the delivery of refined petroleum products, an installation without an air eliminator is considered to meet the intent of section 245, and is considered to be an effective automatic means to minimize the passage of air or vapour through the meter, if it complies with all of the conditions listed in this policy.

5.0 Policy

5.1 The storage tank is above ground.

5.2 Means are provided to ensure that the level of liquid in the storage tank is such that no air or vapour can be drawn into the piping to the measuring system, and that the delivery is inhibited and cannot be initiated unless the tank contains sufficient product. These means may consist of:

- low level sensors interlocked to the pump, and/or
- an automatic tank gauging system, and/or
- a terminal automation system which monitors inventory in real-time and has automatic daily reconciliation against product receipts and

sales, and which is further backed by manual tank gauging.

5.3 The pump is installed so that no section of its suction piping exceeds the elevation of the controlled minimum level of liquid in the tank.

5.4 The meter is at a lower elevation than the controlled minimum level of liquid in the tank.

5.5 The pump supplying the meter is a non-self-priming centrifugal pump.

5.6 The pump is installed so that there is no possibility of product vaporization at the pump inlet; that is, the pump inlet pressure is not less than the net positive suction head for that pump, when the storage tank is at its minimum level and at any pump operating condition.

5.7 The pump is equipped at its inlet with a tap and valve to accommodate a pressure/vacuum gauge during inspection.

5.8 If the installation necessitates piping bridges, these bridges are equipped with manual high point bleed valves for system priming.

5.9 The piping is installed so that there is no possibility of air leaks if a vacuum condition occurs.

5.10 The installation has an automatic control valve installed downstream of the meter, which opens only after the pump has been energized and sufficient time has elapsed for any vapour to be compressed into liquid (10 seconds minimum).

5.11 Where the installation contains control or automatic valves, the sequence of valve openings begins at the control valve nearest to the storage tank and ends at the control valve downstream of the meter.

5.12 There is no common piping between the installation intended for the delivery of product through the meter and the installation intended for the receipt of product into the storage tank.

6.0 Revision

The original bulletin V-21 was issued April 4, 2001. This bulletin was created from a system of rulings and interpretations and has been effective since July 01, 1990.

6.1 The purpose of revision 1 was to revise the bulletin format and make minor editorial changes to suit the format where such headings as a purpose and scope were included.

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